

Growth and productive efficiency of university intellectual property licensing

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Abstract

Licensing activity of US universities has increased substantially in recent years. We examine this increase focusing on the ‘productivity’ and ‘changes in productivity’ of licensing activity. While it is generally acknowledged that there has been a dramatic increase in licensing, there is little understanding of the licensing process as it relates to university characteristics, nor is there evidence on the extent to which this is the result of increased resources devoted to commercialization or whether it stems from other factors. This paper employs data envelopment analysis (DEA) combined with regression analysis to examine the overall productivity of university licensing activity as well as the productivity of individual universities. We find that licensing has increased for reasons other than increases in overall university resources. Given input levels, universities are today more commercially productive than they were in the recent past; we propose reasons for this shift and analyze factors associated with the shift and factors associated with relative levels of commercialization. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

It has been suggested in a number of venues that university resources are not fully exploited as a source of economic growth and competitiveness and recent public policy has been aimed at increasing the commercial impact of universities.¹ For reasons of greater US ‘competitiveness’, the federal government has encouraged greater interactions between universities and the private sector. The Bayh–Dole Act of

1980 changed the nature of ownership of inventions developed under federally funded programs. With passage of the Act, universities could elect to retain title to such inventions, but they are required to file patent applications on those inventions. The Act also encourages the technology transfer activities of universities. The federal government has not been alone in encouraging university licensing activity. State governments have looked increasingly toward public universities as sources of economic development.²

Have these efforts made a difference? According to results reported in the AUTM Licensing Survey (Association of University Technology Managers, 1997), as well as AUTM press releases, university

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¹ See for example, Grey et al. (1986); Public Policy Center for Stanford Research International (1986); Geisler and Rubenstein (1989); National Academy of Sciences (1992); National Science Board (1993) and Grossman et al. (2001).

² See for example, Parnes et al. (2000).

licensing activity has increased over the 6-year period during which AUTM has collected data. Using the sample of universities that participated in all years of the survey, we find that the number of license agreements rose by nearly 70% and the amount of royalties received (in real terms) more than doubled between 1991 and 1996. Prior to 1980, fewer than 250 patents were awarded annually to universities; currently, over 1500 patents are awarded annually. The number of universities actively engaged in technology transfer has increased eight-fold since 1980 to now more than 200.

In this paper, we examine closely this increase in licensing activity. In particular, our focus will be on the ‘productivity’ and ‘changes in productivity’ of licensing activity by US universities over a 6-year period. While it is generally acknowledged that there has been a dramatic increase in university licensing and patenting, there is little understanding of the licensing process as it relates to university characteristics, nor is there evidence on the extent to which this is the result of increased resources devoted to commercialization or whether it stems from other factors. This paper employs data envelopment analysis (DEA) combined with regression analysis to examine the overall productivity of university licensing activity as well as the productivity of individual universities. We find that licensing activity has increased for reasons other than increases in overall university resources. That is, given input levels, universities are today more commercially productive than they were in the recent past; we propose reasons for this change. The regression analysis allows us to relate efficiency to university characteristics. We find that private universities tend to be more efficient in commercialization than public, while universities with medical school are less likely to be efficient. The latter result is particularly interesting given that the majority of university licenses are in the life sciences. We also examine efficiency as related to the size and quality of research faculty in biological sciences, engineering, and physical sciences. Our measures of faculty input in biological sciences and engineering are significantly related to efficiency, while they are not for physical sciences; we propose reasons for this difference.

In our study of the licensing of university intellectual property (IP) we consider the levels and changes in the levels of the ‘outputs’.

1. Sponsored research agreements between universities and industry.
2. License agreements which permit the use of university IP by private sector firms.
3. Royalty payments received by universities in exchange for the use of IP.
4. Disclosures by faculty to their central administration of potentially commercializable innovations.
5. University patent applications.

Each of these outputs is an integral part of a university’s efforts to obtain commercial rewards from the creation of IP.³ We settle on these five outcomes as commercialization outputs based on interviews and survey results reported in Thursby et al. (2001). Each is viewed as important in measuring the success of a university’s technology transfer. For want of better terms we use ‘commercialization’, ‘commercial activity’ and ‘licensing activity’ as succinct ways of referring to the process by which universities produce these five outcomes.

2. Relevance to policy

Whether in response to government incentives for commercial activity or whether it has followed from greater university needs/wants for revenues, the evidence suggests a shift in university goals toward greater commercialization. The Public Policy Center for Stanford Research International (1986) reported in that more than 90% of universities in their sample intended to increase interactions with industry. Morgan et al. (1997) report said that survey results which show increasing responsiveness of engineering faculty to industry needs. Lee (1996) reports a similar finding based on his survey of engineering and science faculty. Cohen et al. (1998) discuss efforts by industry and policy makers to make university research more relevant to industry, and they report growing ties between universities and industry. Finally, we note that university

³ We do not include start-up companies as output. AUTM defines start-ups as new firms that are formed using a license from the university. As such, we do not view start-ups as output that is necessarily different from a license to an existing firm which then starts a new division or produces a new product based on that license.

‘research parks’ designed, in part, to facilitate commercialization of university technology are increasing.

Changing university goals has been accompanied by industry looking increasingly to universities for new technologies. Roessner and Wise (1994) and Thursby and Thursby (2000a) report increasing industry participation in university research.

Have the Bayh–Dole Act and state and federal government efforts to increase commercialization made a difference? Have changing university and industry interests made a difference? The record suggests yes; both AUTM and the US General Accounting Office (1998) have concluded that substantial changes have taken place in university licensing efforts and successes. Mowery et al. (2001) find modest effects of Bayh–Dole on research content, but they find considerable effects of Bayh–Dole on marketing. While the record suggests that the answers are yes, little has been done either on the relative productivity of universities with regard to commercial activities or on the relationship between university characteristics and commercial activity.

Finally, this increase in university commercial activity is not without its critics. Barfield (1998) suggests that universities are inappropriately straying from the public goods justification of federally funded R&D and this has followed, in part, from unintended consequences of the Bayh–Dole Act.

In this paper we look at relative university commercial productivity and factors related to levels of and changes in that productivity. In Section 3 we state more clearly the problem. Following that section we discuss the primary method of analysis, the data, and then present our detailed results.

3. The problem

Research creates IP, and it is our purpose to study the creation and management of that portion of IP with commercial potential. We have three goals. First, we wish to measure relative productivity or efficiency of individual universities in the licensing of their IP. Second, we seek to shed light on factors associated with different commercialization success across universities. Finally, we show that university licensing activities have expanded substantially in the recent past, and we address whether that expansion has

followed from a ‘catching’ up by lagging institutions or from expanding activity of all institutions.

Commercial output can vary over time and across universities, either because of changes in the amount of resources devoted to the creation and licensing of commercializable IP, or because of changes which make resources more productive in those activities. There are two primary problems we face. First, there are multiple measures both of university licensing output and of the inputs used to produce that output. These output and input measures cannot be easily aggregated due to the lack of prices for some outputs and inputs. We solve this technical problem by using DEA which is a linear programming approach to aggregating outputs and inputs and measuring productive efficiency. Second, and of substantially greater difficulty, is the fact that the inputs to university commercial activity are also inputs to teaching, service, and non-commercializable research (i.e. basic research). While we can measure the number of faculty engaged in research and the funds available with which to conduct that research, we cannot measure the intensity of efforts directed toward teaching, basic research, etc. That is, we cannot distinguish an increase in commercial output that follows from a reallocation of inputs from, say, basic to applied (commercializable) research from an increase in output that follows from technical change.⁴

Before proceeding, it should be emphasized that the terms productivity, success and efficiency must be used guardedly as these terms might imply that an inefficient university is poorly run. This is not necessarily the case. A university might specialize in basic over applied research, or it may have a greater preference for teaching over research vis-à-vis other institutions. If such specialization or preferences hold for a university, then it might fare poorly on measures of commercialization success because the showing follows from specialization and preferences and not from competencies.

University commercialization can be characterized by the following. Research is conducted by faculty, though it is not necessarily conducted with the intent of creating a commercializable innovation or invention. This research may or may not have been sponsored

⁴ Technical change is here used in the standard sense of any change in output that is unrelated to changes in inputs.

by the private sector (industry sponsored research). If a faculty member believes that results of research are commercializable he or she undertakes a formal, confidential process of disclosure of the results to the university's transfer technology office (TTO) which evaluates the innovation for patentable and commercial potential. If deemed commercializable, the TTO might seek patent protection and does seek to find private sector firms as licensees of the technology. If licensed, the university might receive royalties at the time the license is executed or over the life of the license. They might also seek sponsored research funds from an interested firm for further development of the IP or for undirected research; this further research may lead to other commercializable technologies.

While it might seem that the ultimate objective of licensing activity is money through either sponsored research or royalties, this is not necessarily the case for every university. University obligations are complex as universities, particularly public universities, are charged with the creation and dissemination of knowledge regardless of whether such activities are currently profitable in the private sector. Many universities view not only royalties and sponsored research as outputs of their commercialization activities, but also the number of patents and disclosures (measures of commercialization efforts) as well as the number of licenses (a measure of the dissemination of knowledge via new or improved products and processes). This is not to say that each of these outputs is equally important (they are not, see Jensen and Thursby, 1999), but the fact that they are viewed as outputs by at least some universities makes the process of measuring commercialization activities difficult.

4. Data envelopment analysis and efficiency measures

Economists say that a producing unit is 'technically inefficient' if it is possible to produce more output with the current level of inputs or, equivalently, it is possible to produce the same output with fewer inputs. Here, we examine the technical efficiency of universities with regard to their licensing activities. We say that a university is technically inefficient (or, simply, inefficient) if, when compared to universities with similar amounts of inputs, it could produce greater commercial output

without increasing its input usage, or, equivalently, it is one which, when compared to universities with similar levels of outputs, could produce the current levels of outputs with fewer inputs. Reasons for technical inefficiency include, among other things, the failure to take advantage of all commercializable IP as well as a greater preference for basic over applied research.

As noted earlier, DEA is a commonly used method to evaluate relative efficiency when there are multiple outputs and when meaningful aggregation is not possible. DEA produces an efficiency rating or score for each university by first determining the set of universities which exhibit 'best practice' with regard to licensing activities. These universities are said to form the frontier that relates outputs and inputs. All other universities are then compared to the subset of these best practice universities they most resemble in terms of levels of outputs and inputs. Thus, for each university in our sample, DEA determines whether it lies on the frontier (exhibits best practice) or, if not, how 'far' from the frontier it lies. Units that lie on the surface are termed efficient and those not on the surface are said to be inefficient.

What is meant by a 'best practice' university? Consider a simple example of a single licensing output and a single input for each of six universities whose outputs and inputs are depicted in Fig. 1. The line linking universities 1–4 is the best practice frontier. From the standpoint of efficiency, no one of these universities dominates the others; each successively uses more input and produces more output. Universities 5 and 6, on the other hand, are dominated by others; for example, university 2 uses less input and produces

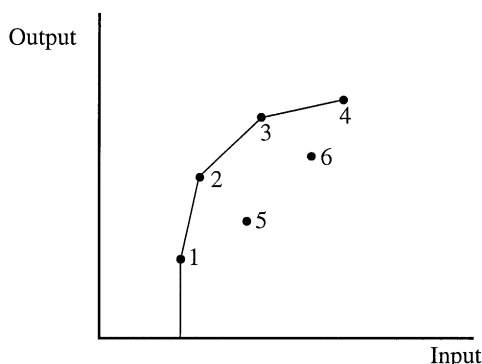


Fig. 1. DEA production frontier.

more output than does 5. Both 5 and 6 lie below the efficiency frontier, and to measure inefficiency they are compared to the nearest facets linking efficient universities. For example, in considering how much less input university 5 could use to produce its current output, we compare it to a ‘virtual’ university that is a convex combination of the outputs and inputs of universities 1 and 2. The distance of 5 from that virtual university is the measure of the inefficiency of 5.

While we say that a university which lies on the frontier is efficient, what we have really determined is only that we do not observe another university that does better given the mixes of outputs and inputs. Further, inefficient universities may be deemed inefficient simply because of preferences for other activities over commercialization. The actual level of input usage in commercialization activity is not necessarily equal to our measured inputs since many of our inputs are used in conjunction with outputs not associated with commercialization (teaching, basic research, etc.).

Care should be used with the terms outputs and inputs as they can appear misleading; for example, industry sponsored research is both an output and an input. Universities use resources to attract industry funding, but that funding is then an input to further research output and, possibly, disclosures, patent applications, etc. Sponsored research is, in part, an ‘intermediate good’ as well as a ‘final product’.⁵ For the purposes of DEA, we think of an output as simply something of which the university wants more and inputs are those resources that are used in getting more of the outputs. Commercialization outputs reflect the licensing activities of a university, while inputs consist of those resources that are important in producing the outputs, but that are not the measures of commercial activity. For the interested reader, a more complete treatment of DEA is found in Appendix A.

5. Data

The AUTM licensing survey has data on the technology transfer programs of a variety of institutions.

⁵ Industry sponsored research may have elements of altruism. However, it is the case that sponsored research agreements are generally for further development of a licensed technology or they include options for the grantor firm to license any innovation resulting from the sponsored research (Thursby and Thursby, 2000b).

We restrict attention to US universities for 1991–1996. The survey is the source for output information: the number of licenses executed (LICENSES), the amount of industry sponsored research (INDSUPPORT), the number of new patent applications (PATENTS),⁶ the number of invention disclosures (DISCLOSURES), and the amount of royalties received (ROYALTIES). The survey provides two input measures: federal support (FEDERALSUP) and the number of professionals employed in the TTO (TTOSIZE). It is clear that the latter is important for commercial activity, but the role of federal support is not as clear. It has been suggested that industry money tends to flow to universities with large federal support; federal support may leverage industry support thereby reducing industry research costs. Further, federally funded research is frequently the source of commercializable IP (US General Accounting Office, 1998).

Other input data are from the National Research Council’s 1992 survey of all Ph.D. granting departments in the US (NRC, 1995). Information is only provided for departments that grant the Ph.D. It is plausible that substantial research programs have difficulty existing in the sciences and engineering—the departments from which the bulk of commercial activity originate (see Jensen and Thursby, 1999)—without Ph.D. students. This implies the reasonable proposition that science and engineering departments that do not grant the Ph.D. are not strong research departments and, hence, provide less inventive input to a university’s commercial activities; the AUTM data supports this proposition. The NRC input data are the number of faculty in each of the three major program areas biological sciences, engineering and physical sciences as well as a faculty quality rating obtained from the NRC’s 1992 peer evaluation survey. Input variables are FACULTY(*i*) and QUALITY(*i*) for total faculty and faculty quality rating in the Ph.D. granting departments of program area *i* (*i* = BIO.SCI for biological sciences; *i* = ENG for engineering and *i* = PHY.SCI for physical sciences).

Initial computations are based on a sample of 112 institutions. For these computations we use averages of the annual values reported in the AUTM

⁶ We use new patent applications rather than patents awarded due to the often substantial lag between patent application and the granting of the patent.

Table 1
Output and input mean values

Outputs	Mean	Inputs	Mean
INDUSTRIESUP ^a	10.83	TTOSIZE	2.50
ROYALTIES ^a	1.91	FEDERALSUP ^a	90.81
DISCLOSURES	53.43	FACULTY (BIO.SCI)	151.79
LICENSES	15.05	FACULTY (ENG)	83.41
PATENTS	16.85	FACULTY (PHY.SCI)	119.99
		QUALITY (BIO.SCI)	2.59
		QUALITY (ENG)	2.20
		QUALITY (PHY.SCI)	2.63

^a Millions of 1996 dollars.

survey. Not every university participated each year so our measures are the mean values in the sample. INDUSTRIESUP, FEDERALSUP, and ROYALTIES are measured in millions of constant, 1996 dollars and QUALITY(*i*) are measured on a scale of 0–5 with 5 indicating a distinguished program. Entries in Table 1 are means averaged over all 112 institutions and over 6 years of the survey. After consideration of the full sample, we compute productivity changes for the 57 institutions that responded in each year of the AUTM survey.

A weakness in our analysis is that some actions that lead to current output may be actions that originated in earlier years. For example, current royalties are based upon research conducted in the past and, to some extent, on licenses executed in the past. We are mixing inputs and outputs from different points in time. To the extent that there are lags involved, our results must be considered with caution. However, it is the case that for some of the data the year to year variation is small. This is certainly true for faculty size and quality and is true to some extent for sponsored research. Also, results for the full set of universities are based on 6-year averages of the data and this should reduce error from time lags.

6. DEA results

6.1. Full sample efficiency scores

The list of all 112 universities and their DEA efficiency scores are found in Table 2. The scores in Table 2 are scaled such that an efficient university has a score of one, and a score greater than one indicates

Table 2
Efficiency scores

University	Score
Akron	1
Ala./Birmingham	1
Ala./Huntsville	1.274
Arizona	1.506
Arizona State	1.927
Ark./Fayetteville	1.335
Auburn	2.070
Ball State	1
Boston	1
Brandeis	1.058
Brigham Young	1
Brown	2.049
CALTECH	1
Carnegie Mellon	1
Case Western	1.795
Central Florida	1
Chicago	1
Cincinnati	1.631
Clemson	1.015
Colorado	1.639
Colorado State	1
Columbia	1
Connecticut	1.739
Cornell	1
Dartmouth College	1.565
Dayton	1
Delaware	1.873
Denver	1
Drexel	1.003
Duke	1
Emory	1
Florida	1.088
Florida Atlantic	1
Florida State	1
Georgetown	1
Georgia	1
Georgia Technology	1
Harvard	1
Hawaii	1.736
IL/Chicago	2.710
IL/Urbana-Cham	1
Illinois Institute of Technology	1
Illinois State	1
Indiana	1
Iowa	1.543
Iowa State	1
Johns Hopkins	1
Kansas	1.709
Kansas State	1.466
Kentucky	1
Louisiana State	1.075
Maine	1
MD/Baltimore	1

Table 2 (Continued)

University	Score
MD/College Park	1.340
Miami	1.647
Miami	1
Michigan	1.585
Michigan State	1
Michigan Technology	1.065
Minnesota	1.144
Mississippi State	2.165
MIT	1
Montana State	1.247
NC/Chapel Hill	1.183
Nebraska-Lincoln	1.621
New Hampshire	1
New Mexico State	1.650
New Orleans	1
NJ Institute of Technology	1
North Carolina State	1
North Dakota State	1
Northeastern	1
Northern Illinois	1
Northwestern	1.418
Ohio	1
Ohio State	1.560
Oregon	1
Oregon State	2.288
Penn State	1
Pennsylvania	1.086
Pittsburgh	1.055
Princeton	1
Purdue	1.019
Rhode Island	1.189
Rice	1.425
Rochester	1.295
Rutgers	1.290
South Alabama	1
South Carolina	2.463
South Florida	1
Southern California	1.656
Stanford	1
Stevens Institute of Technology	1.912
Syracuse	2.882
Temple	1
Tennessee	1.403
Texas A&M	1.095
Texas at Austin	1.527
Tufts	1.068
Tulane	1
Tulsa	1
Utah	1
Vanderbilt	1.028
Virginia	1.012
Virginia Technology	1.140
Wake Forest	1
University of Washington	1.302

Table 2 (Continued)

University	Score
Washington University	1
Washington State	2.020
Wayne State	1.645
Wisconsin-Madison	1.170
Yale	1.101

the proportionate increase in commercial output that could be produced if the university were to move to the frontier (become efficient). For example, Akron has an efficiency score of one, hence we have no evidence that it is inefficient. On the other hand, Alabama at Huntsville's score of 1.274 suggests that, based on its level of inputs and the performance of the efficient universities, it could increase its commercial output by more than 25%. Fifty-eight universities exhibit some degree of inefficiency. Bunching of scores at the efficient value is common in DEA analysis.

We conduct two analyses of the efficiency scores. We begin with two-way tabulations of inputs or outputs and whether a university is efficient. We then regress efficiency scores on the inputs and outputs. That analysis will allow us to consider effects on efficiency of changes in each of the inputs and outputs, holding constant the values of other inputs and outputs. Rather than use the scores in Table 2, in our analyses we use an indicator variable for whether or not the university is efficient. This is done because, in a DEA analysis, one has greater confidence in the separation of universities into efficient versus inefficient than in the actual scores of the inefficient universities (see Appendix A). The indicator variable is IEFF where $IEFF = 1$ if the university is efficient and $IEFF = 0$ if the university is inefficient.

6.2. Two-way comparisons

We divide inputs and outputs (approximately) into quintiles, and then count the number of efficient or inefficient universities in each quintile. Pearson χ^2 tests are used to test for association. To reduce the number of tables to a manageable number we use the total number of faculty in each university (FACULTY), and the weighted average quality rating (QUALITY) where the weights are faculty size. As an added measure of a university's intensity of licensing efforts we

form a new variable, TTOSIZE/FAC, which is the ratio of TTOSIZE per 100 faculty.

We consider two environmental variables that could influence efficiency. The first is whether a university has a medical school (MEDSCH = 1 if it has a medical school, 0 otherwise), and the second indicates whether the university is private (PRIVATE = 1 if private, 0 otherwise). We hypothesize that universities with medical schools will be more commercially efficient since a large fraction of university licenses are in the biomedical area, and, since clinical trials are conducted in hospitals, universities with medical schools are more easily able to engage in late stage development of such products. We also hypothesize that private universities are more efficient as they are able to specialize to a greater extent than are public universities which have greater service commitments and often more substantial teaching duties. As our sample is restricted to research universities, we may have a sample of private universities that value research highly, and, if they specialize, may devote fewer resources to teaching and service.

Table 3 presents the contingency tables that relate whether a university is efficient to five outputs (INDUSTRYSUP, ROYALTIES, DISCLOSURES, PATENTS and LICENSES), five inputs (FEDERALSUP, FACULTY, QUALITY, TTOSIZE and TTOSIZE/FAC) and two environmental variables (MEDSCH and PRIVATE). The values in the first column of each of the 12 contingency tables are bounds for the 10 inputs and outputs and values for the two environmental variables. Entries in the remaining two columns are the number of universities that are efficient (IEFF = 1) or inefficient (IEFF = 0). For each table we include the P -values for the χ^2 test of independence between efficiency and the input or output. For example, the tabulation of IEFF versus ROYALTIES reveals that, among universities with \$320,000–750,000 of royalty income, 16 are inefficient and six efficient, and the test of independence between ROYALTIES and IEFF is significant at a level of 3.2% or smaller.

With the exception of PATENTS and LICENSES, we reject the hypothesis of independence between IEFF and the outputs at a significance level of 15% or lower. The count pattern for INDUSTRYSUP, ROYALTIES, and DISCLOSURES suggests that school in the largest and smallest quintiles are more likely to be efficient than are schools in the three middle quintiles.

Efficiency is significantly related to each of the inputs. Note that, with the exception of TTOSIZE/FAC, it is schools with the smallest amounts of inputs that are most likely to be efficient. We find interesting the suggestion that schools with the lowest quality rating are the most likely to be efficient. This may be due to tendencies for the highest quality faculty to specialize in basic research which, in general, can result in less inventive activity.⁷ When we consider the number of technology managers per 100 faculty (TTOSIZE/FAC), we find that it is the schools in the largest quintile which tend to be efficient; to the extent that this measures a university's relative intensity of effort for commercialization, this result is not surprising. Comparing efficiency and environmental variables reveals a significant relationship only between IEFF and PRIVATE where, as expected, we find that a greater likelihood of efficiency for private schools.

The Table 3 results must be qualified since nothing is held constant in a two-way comparison. Failure to control for other effects can mask relationships, and it can suggest relationships between variables when the variables are unrelated, but where each is related to a third variable. For example, we note above that FACULTY and QUALITY are related to efficiency. Since these two variables have a moderately high correlation (0.72), it may be that, for example, only FACULTY is actually related to efficiency but QUALITY appears to be related to efficiency simply because of its correlation with FACULTY, a correlation that may have nothing to do with efficiency. This need to measure effects holding constant levels of other factors leads to regression analysis.

6.3. Regression results

We consider logit regressions with the indicator variable IEFF as the dependent variable. Table 4 presents results for IEFF considered first in a regression with outputs and inputs only and then in a regression which includes the environmental variables. In addition to coefficients and standard errors, Table 4

⁷ On a related note, Mansfield and Lee (1996) report that industry sponsored basic research tends to be biased toward higher quality departments, while there is no bias in industry sponsored applied research.

Table 3
Contingency tables

IEFF	0	1		0	1
INDUSTRIESUP			FEDERALSUP		
≤3.25	5	15	≤19.5	4	18
3.25–5.5	15	8	19.5–46	17	6
5.5–9.75	13	10	46–83	13	8
9.75–17	14	8	83–140	11	12
>17	11	13	>140	13	10
P-value		0.056	P-value		0.003
ROYALTIES			FACULTY		
≤0.1	6	14	≤100	3	17
0.1–0.32	14	9	100–235	14	11
0.32–0.75	16	6	235–350	12	8
0.75–2	13	10	350–550	13	10
>2	9	15	>575	16	8
P-value		0.032	P-value		0.008
DISCLOSURES			QUALITY		
≤13	8	14	≤2.1	6	17
13–28	14	9	2.1–2.55	10	10
28–45	12	8	2.55–3	16	6
45–82	15	8	3–3.5	15	8
>82	9	15	>3.5	11	13
P-value		0.135	P-value		0.018
PATENTS			TTOSIZE		
≤4.5	10	12	≤0.92	7	16
4.5–9	12	10	0.92–1.4	15	6
9–14.75	15	9	1.4–2	10	11
14.75–25	11	9	2–3.5	16	9
>25	10	14	>3.5	10	12
P-value		0.628	P-value		0.051
LICENSES			TTOSIZE/FAC		
≤2.75	8	13	≤0.325	12	10
2.75–5.5	13	10	0.325–0.55	16	7
5.5–11.5	13	9	0.55–0.8	13	9
11.5–22	13	9	0.8–1.3	11	9
>22	11	13	>1.3	6	19
P-value		0.547	P-value		0.024
MEDSCH			PRIVATE		
0	26	30	0	42	31
1	32	24	1	16	23
P-value		0.257	P-value	0	1

includes, for continuous variables, the elasticities of the probability that the dependent variable equals one with respect to each of the regressors. For the indicator variables MEDSCH and PRIVATE the odds ratios are presented. The elasticities give the percent change in the probability that a university will become efficient if the value of an input or output were to rise by 1%. For example, the value of 0.815 for INDSUPPORT

in the first IEFF column means that a 1% increase in industry support will increase the probability by 0.815% that a university is efficient. Elasticities allow a comparison of the importance of various outputs and inputs in determining efficiency. For MEDSCH and PRIVATE we present the odds ratio for a change in MEDSCH or PRIVATE from 0 to 1. For example, the value of 4.438 for PRIVATE in the second IEFF

Table 4
Logit regression results

	Coefficient	Standard error	Elasticity	Coefficient	Standard error	Elasticity/ odds ratio
INDUSTRYSUP	0.152	0.059	0.815	0.208	0.063	1.119
ROYALTIES	0.242	0.107	0.156	0.292	0.113	0.189
DISCLOSURES	0.015	0.028	0.378	0.030	0.026	0.735
PATENTS	0.136	0.089	0.878	0.135	0.083	0.871
LICENSES	0.167	0.073	1.273	0.165	0.067	1.257
TTOSIZE	−1.271	0.477	−1.642	−1.401	0.540	−1.810
FEDERALSUP	−0.007	0.007	−0.327	−0.009	0.007	−0.445
FACULTY (BIO.SCI)	−0.007	0.004	−0.662	−0.005	0.004	−0.508
FACULTY (ENG)	−0.013	0.011	−0.657	−0.021	0.012	−1.076
FACULTY (PHY.SCI)	−0.006	0.009	−0.435	0.002	0.011	0.128
QUALITY (BIO.SCI)	−0.467	0.456	−0.664	−0.459	0.446	−0.654
QUALITY (ENG)	−1.435	0.577	−1.940	−1.560	0.662	−2.109
QUALITY (PHY.SCI)	−0.104	0.638	−0.150	−0.555	0.778	−0.802
MEDSCH				−1.867	0.966	0.145
PRIVATE				1.490	0.803	4.438
R^2	0.482			0.530		
Number of observation	112			112		

column implies that the probability of efficiency for a private school is more than four times that of a public school, all other factors held constant.

In interpreting results bear in mind that the elasticities are partial effects holding constant *all* other inputs and outputs. Thus, each output should have a positive elasticity since, holding constant all inputs, an increase in an output is clearly a move toward efficiency. Likewise, input elasticities should be negative as an increase in an input, holding output constant, is a move away from efficiency.

In the regression without the environmental variables, all output elasticities are positive, and, with the exception of DISCLOSURES, each associated coefficient is significant. LICENSES, with an elasticity of 1.273, has the greatest impact on efficiency scores. The importance of licenses executed follows from its central role in the commercialization process. Faculty disclose IP that they wish to have licensed, patents are to protect IP that is licensed or believed licensable, royalties flow from licensed IP, and industry support is either for further development of a licensed technology or the sponsoring firm retains some rights to license any commercializable IP.

The smallest elasticity is the one associated with royalties. We believe this stems from two factors. First, royalties are highly skewed among universities

and tend to be associated with only a few licenses. In Jensen and Thursby (1999) it is reported that, on average, 78% of royalty income comes from only five licenses so that substantial royalty income from a specific license has a very low probability. Second, some universities are willing to forego royalty income for sponsored research funds; in other words, preferences for royalty income are highly varied across universities.

The insignificant DISCLOSURES coefficient can be taken to imply a substantial number of inefficient universities with large numbers of invention disclosures and/or a substantial number of efficient universities with small numbers of invention disclosures. It is perhaps not surprising that invention disclosures might have greater variation across institutions regardless of efficiency. Of the five outputs, only DISCLOSURES and PATENTS are entirely within the discretion of university personnel (the others require agreements with industry), and DISCLOSURES have low cost compared to PATENTS (the patent process can cost from 15,000 to 20,000 dollars). Further, some universities informally evaluate or pre-screen new technologies with regard to commercial appeal; those judged likely to have such appeal go through the formal disclosure process. Pre-screening by a subset of universities implies that DISCLOSURES is

counted differently at those universities compared to those that do not pre-screen.

All inputs have the expected negative elasticities, but only the number of technology transfer professionals (TTOSIZE), the number of biological science faculty (FACULTY (BIO.SCI)), and the quality of the engineering faculty (QUALITY (ENG)) are significant. Insignificance of so many inputs is probably due to their use in production of other university outputs. To the extent that this is true, our input measures are imperfect measures of the effort expended in licensing activity. The largest elasticities are for QUALITY (ENG) (−1.94) and TTOSIZE (−1.642). TTOSIZE is the only input solely associated with licensing activity so that a large elasticity is expected. Engineering is more applied than the biological and physical sciences, hence it is not surprising to find it to be important in creating commercializable IP, but we are surprised at the magnitude of its elasticity and the apparent fact that the number of engineering faculty is of less importance. The opposite is found with biological sciences for whom quality appears not to be important but the number of faculty is important. It has been noted to us that the market for university biological science IP has tended to be a seller's market, whereas the market for engineering and physical science IP has tended to be a buyer's market.⁸ If this is true, then universities should do better in the biological sciences in finding licensees and extracting rents. Further, in interviews with licensing professionals several interviewees volunteered that researchers in biotechnology tend to have a culture that is more encouraging of commercial activity than is the case in the physical sciences. Data collected by AUTM support these differences in markets and cultures. In 1996, AUTM collected license information split by life sciences versus physical sciences. For our sample, aggregate royalties from life science licenses is around four times that of the physical sciences. Further, royalties per active life science license is around 2.5 times that of the physical sciences. The relative importance of FACULTY (BIO.SCI) is in keeping with this difference in markets and cultures, but the quality results are not. Our finding of importance only

of total faculty in the biological sciences and quality of the engineering faculty merits further investigation.

Finally, the coefficient of federal support is insignificant. This is surprising given that the Bayh–Dole Act encourages commercial activity related to federal research money. We suggest that the insignificance of FEDERALSUP may come from the preponderance of federal support directed toward basic research; as such it would be hard to detect a relation between commercialization and federal support.

In the last three columns of Table 4 are elasticities and odds ratios when the environmental variables are included as regressors. Both environmental variables are significant at a 10% level. Recall that our earlier analysis showed only a relation between PRIVATE and efficiency. Here, we find that universities with medical schools are less likely to be efficient and private schools are more likely to be efficient. The result on private schools is expected. We are surprised at the medical school results given the ease of conducting clinical trials when a medical school is present and the fact that a large fraction of university licenses are related to biomedical inventions. It may well be that the heavy service commitments of medical schools serves to reduce commercialization efficiency. All output and input elasticities remain similar to their values in the first regression with the exception that the number of engineering faculty is significant with a fairly large elasticity and FEDERALSUP is significant, albeit with a small elasticity.⁹

6.4. Year to year productivity changes

Year to year efficiency changes can be examined for universities that appear in each of the six survey years. There are 57 such universities. In our first experiment we include each of the 6 years and each of the 57 universities in a single DEA analysis. This implicitly assumes there are 342 universities (57 schools by 6 years). We are looking for 'best practice' across 6

⁸ We are indebted to Dan Massing of Cornell research foundation for this interpretation.

⁹ We also considered as regressors the split of research money across 17 science and engineering departments/schools. The notion is that a more disaggregated split of research effort than the one we use (three major program areas) might shed light on commercialization efficiency. Both the aggregate research levels and proportions were considered, the coefficients are not jointly significant different from zero. For the sake of brevity, we do not present the detailed results.

Table 5
Year to year efficiency comparisons

Year	Number inefficient	Average score	Potential percent of increase in outputs				
			INDUSTRYSUP	ROYALTIES	DISCLOSURES	LICENSES	PATENTS
1991	48	1.737	57.2	45.5	56.5	53.3	57.9
1992	40	1.574	42.1	28.8	37.5	33.4	38.9
1993	45	1.514	40.4	25.5	39.5	37.0	39.3
1994	40	1.433	28.0	17.1	30.8	27.9	30.3
1995	37	1.413	25.1	14.3	26.0	21.3	23.6
1996	28	1.366	19.9	9.4	19.9	14.9	18.8

years and 57 universities. A count of the number of efficient institutions by year and averages of the scores by year provides information on whether efficiency has changed over time. We also use the scores to compute the potential outputs that would have occurred if each university had been efficient. If the efficiency score is ϕ and the actual level of some output is A , then potential output is $A\phi$. We then compare aggregate potential outputs to actual outputs.

Table 5 presents, for each year, the number of efficient universities, the average efficiency score and the percentage increase in each of the five outputs that would be expected under efficiency. The evidence clearly indicates increasing efficiency over time. Universities in each year are generally closer to the frontier than they were in the preceding year. The change in efficiency is particularly striking when one considers the shortfall from potential output. In 1991, the universities could have been producing about 55% more commercial output had each university been efficient, whereas in 1996 they could only have produced about 17% more output. Note that the potential gain in royalties is smaller in each year than it is for any other output. This suggests that inefficient universities tend to be closer to efficient universities in royalties received than in any other output dimension; inefficient universities do relatively 'best' with royalties received.

The results in Table 5 suggest productivity improvements over time, but they are only suggestive of the extent to which the frontier has shifted over time (technical change), and the extent to which inefficient universities are 'catching up' to the efficient universities. To address these issues we use DEA linear programs and a Malmquist total factor productivity (TFP) index to measure changes in productivity (see Fare et al., 1994). The Malmquist TFP index measures the

change in the productivity of the inputs at two points in time. A value of the index >1 indicates TFP growth between two periods; an index value of, say, 1.1 implies 10% factor productivity growth. As we have a 6 year panel we calculate five values of the index for each university (one for each pair of adjoining years). For a formal statement of the index, see Appendix B.

The geometric mean for all universities is 1.079, which indicates an average annual growth in TFP of 7.9% over the period 1991–1996; i.e. if input levels had been constant over this period, there would still have been a growth in commercialization output of 7.9% annually. Only 13 (23%) of the universities do not appear to have experienced productivity growth.

The aggregate figure for TFP can be decomposed into the product of two growth measures: productivity growth due to movements of inefficient universities toward the frontier and expansion of the frontier itself (see Fare et al., 1994). Changes in relative efficiency of inefficient institutions is a sort of 'catching up'. The decomposition suggests that inefficient universities have been catching up to efficient universities at an annual rate of only 0.4%, hence the bulk of the TFP growth follows from expansion of the frontier for which the annual growth rate is calculated to be 7.5%. It would appear that, in general, universities are increasingly involved in commercialization, but this increasing commercialization stems largely from an expansion of the frontier rather than from a catching up by inefficient institutions. The frontier is expanding and most institutions are 'chasing' that frontier.

The standard use of the term technical change refers to a change in output that does not follow from a change in inputs; thus, the expansion of a production frontier is typically regarded as being due to technical change. However, as discussed earlier, it is difficult to

measure the actual inputs used in commercialization because those inputs are used to produce other university outputs. What we are calling technical change may well be a reallocation of inputs from another activity toward commercialization so that, while we might observe constant ‘aggregate’ levels of university inputs, it might be the case that they are being used more intensively in commercialization. We suggest that the expansion of the frontier (technical change) stems from a change in the environment for university commercialization that involves a reallocation of inputs, a change in market demand for university IP, and/or increasingly experienced, knowledgeable and demanding TTOs.

The introduction cites sources that suggest a change in both university preferences for commercialization activities and a change in industry demand for university technologies.¹⁰ If acted upon, the change in university preferences implies a possible shift of resources away from other activities toward activities with potential commercial appeal. This change in preferences might also imply a greater share of commercially viable IP being disclosed by faculty. In interviews with technology transfer officers they were asked about commercializable innovations that are not disclosed. While reluctant to offer numbers, most of them suggested that they would not be surprised if they were seeing substantially less than half of such innovations, though several noted an increasing willingness of faculty (particularly younger faculty) to disclose. The increase in demand for university IP can be expected to improve the matching of university IP with firms that can exploit the commercial potential of the IP. University technology transfer officers suggest that a major problem in licensing a technology is simply finding a firm that is suited for and interested in the technology; it is sufficiently difficult that bidding for a technology is rare (Jensen and Thursby, 1999). Increased industry interests in university IP serves to improve this matching process both by reducing the costs of searching for a licensee and by increasing the probability of a match. Finally, from interviews with both industry and university licensing personnel, it is clear that university TTOs are increasingly sophisticated and demanding in their license dealings with industry.

7. Conclusion

The federal and state governments have encouraged universities to commercialize the results of their research. Universities espouse an increasing willingness to engage in such activities and private sector firms are looking more carefully at university IP. We examine university commercial activities in order to shed light on the activities of individual universities as well as the overall direction of such activities. Commercial activities include industry sponsored research and royalties as well as numbers of invention disclosures, licenses executed and new patent applications. University productivity or efficiency with regard to such activities varies not only according to the capabilities of the faculty and staff, but also according to university preferences in the use of their resources. We use DEA to measure the relative efficiency of each university in our sample. Both contingency table analyses and regressions are used to relate the efficiency scores to the levels of all commercial outputs as well as to the levels of inputs. We also examine the changes in productivity of university commercial activity over time. Our major results are the following.

- We find substantial evidence of inefficiencies. As we note, this inefficiency may stem simply from university preferences for or specialization in outputs unrelated to licensing activity (such as basic research and teaching) rather than from competencies in licensing.
- There has been substantial growth in commercial activities of US universities. We attribute this both to a changing environment within universities regarding commercial activity as well as an increasing desire of industry for university technologies.
- This growth has stemmed primarily from a growth in commercialization by all universities rather than a ‘catching up’ by the inefficient institutions.
- We find that biological sciences and engineering are more important to licensing activity than are the physical sciences. We attribute this to the more applied nature of engineering and the better market opportunities and orientation toward markets of biological sciences.
- We find that, on a number of dimensions, the smallest schools tend to be more like the largest schools than the mid-range schools.

¹⁰ For more on this, see Thursby and Thursby (2000a).

- The lower the research quality of a university the more efficient the university tends to be in commercial activity. This we interpret as resulting from greater specialization in basic research of the higher quality research faculty.
- We find that private universities are more likely to be efficient and universities with medical schools are less likely to be efficient.

It has been suggested in a number of venues that university resources are not fully exploited as a source of economic growth and competitiveness. Our results clearly show that this criticism is decreasing in its relevance, though the criticisms of those who feel that universities have gone too far in commercialization activities may be increasing in relevance.

What we have not been able to fully determine is the source of the growth in university commercial activity. To what extent is it due to a shift of resources away from basic research to more commercializable research?¹¹ To what extent does it follow from an increasing willingness and ability of universities to license their IP? To what extent is this growth due to a greater demand by industry for university IP? Answers to these questions are the subject of future research.

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Appendix A

DEA is a non-parametric rather than a statistical procedure. As such there are no implicit distributional assumptions. If decision making unit (DMU) i produces y_i units of output with x_i inputs, then it should be the case that any other DMU j should also be able to do at least as well in the following sense. If j has an amount x_i of inputs, then j should be able to produce at least y_i output, otherwise j is inefficient. If j produces more than y_i using x_i amount of inputs, then i is inefficient. If j produces an amount y_i of output then j should be using no more than x_i of inputs, otherwise j is inefficient. If j uses less than x_i of inputs to produce y_i then i is inefficient. Schematically, we have the following factors.

Let i use x_i and produce y_i , then:

(i) if j uses x_i then

$$y_j < y_i \Rightarrow j \text{ is inefficient}$$

$$y_j > y_i \Rightarrow i \text{ is inefficient}$$

$$y_j = y_i \Rightarrow \text{no evidence that } i \text{ or } j \text{ is inefficient;}$$

(ii) if j produces y_i then

$$x_j > x_i \Rightarrow j \text{ is inefficient}$$

$$x_j < x_i \Rightarrow i \text{ is inefficient}$$

$$y_j = y_i \Rightarrow \text{no evidence that } i \text{ or } j \text{ is inefficient}$$

DEA is formulated as a linear program to find the 'best' virtual DMU for each real DMU i . If the virtual DMU produces more outputs with the same inputs or the same outputs with less inputs, then i is inefficient,

¹¹ Wes Cohen suggested to us that data the NSF collects on the split between university basic and applied research might be useful in considering preferences for basic versus applied research. The aggregate figures are presented in 'Science and Engineering Indicators'. We contacted NSF and discovered that, after entering the university data on the split and calculating the aggregate figures, the university level information is discarded. Reasons appear to be both very strict confidentiality agreements made to universities and lack of faith in the individual numbers as there is not a consistent definition or mechanism for determining what is and what is not basic and applied research spending so that, in their opinion, the numbers from school to school are generally not comparable. This is unfortunate. Even if the numbers are not an accurate reflection of the actual split, they do provide evidence on what universities *think* they are doing and a time series might prove useful in the debate over whether faculty have shifted their research focus more toward applied research.

otherwise, we have no evidence that i is inefficient. There are a number of formulations of DEA, but the basic problem can be presented as follows. Let there be n DMUs each with vectors of outputs y_i and vectors of inputs x_i , $i = 1, \dots, n$. To determine productive efficiency of DMU i , DEA involves the solution of the problem

$$\max_{u,v} \frac{u'y_i}{v'x_i}$$

subject to

$$\frac{u'y_j}{v'x_j} \leq 1, \quad j = 1, \dots, n$$

$$u, v \geq 0$$

Since there are an infinite number of solutions to this problem, we add the further constraint $v'x_i = 1$. If no combination of DMUs (virtual DMU) has a larger ratio than DMU i , then i is efficient. This problem is solved for each of the n DMUs. To understand more clearly the DEA approach, consider a simple case where each DMU produces a single output using a single input. In such a situation one can determine efficiency by looking at the ratio of output to input. In the multiple input, multiple output setting, DEA produces a composite or 'virtual' output $u'y_i$ which is compared to the virtual input $v'x_i$.

One can also solve the equivalent envelopment form, following duality in linear programming.

$$\max_{\phi, \lambda} \text{subject to } -\phi y_i + Y\lambda \geq 0,$$

$$x_i - X\lambda \geq 0, \quad \text{and} \quad \lambda \geq 0$$

where Y and X are matrices of outputs and inputs for all universities and $1 \leq \phi \leq \infty$. If $\phi = 1$ we have no evidence that DMU i is inefficient, whereas $1/\phi$ is the fraction of 'best practice' output for the DMU.

Researchers typically use either a constant or variable returns to scale (VRS) model. The VRS model is less restrictive in that it allows for any scale effect, and, for that reason, it is increasingly the choice for researchers. For the VRS model we add the constraint $\mathbf{1}'\lambda = 1$, where $\mathbf{1}$ is an $n \times 1$ vector of ones. A DEA model can have an input or output orientation; in the input (output) orientation the focus is on the optimal reduction (increase) in inputs (outputs) that would

follow if an inefficient DMU were to move to efficiency. Here, we consider an output orientation and the VRS model. The above envelopment form is the output oriented form of DEA. An output and an input orientation produce the same set of efficient universities, however, the scores of the inefficient universities vary according to the orientation. In Section 6 we report the value of ϕ , the efficiency score for each university in our sample. Discussions of DEA can be found in Seiford and Thrall (1990); Charnes et al. (1994); Ali and Seiford (1993) or Fare et al. (1994).

Appendix B

With panel data one can use DEA like programs to measure productivity change over time. Fare et al. (1994) suggest as a measure of TFP change the geometric mean of two Malmquist indices, one of which is based on technology in period t and the other on technology in period $t + 1$, or

$$m(y_{t+1}, x_{t+1}, y_t, x_t) = \left[\frac{d^t(x_{t+1}, y_{t+1})}{d^t(x_t, y_t)} \times \frac{d^{t+1}(x_{t+1}, y_{t+1})}{d^{t+1}(x_t, y_t)} \right]^{1/2}$$

where, for $k = t$ or $k = t + 1$, the distance functions $d(\cdot)$ are defined as

$$[d^k(x_k, y_k)]^{-1} = \max_{\phi, \lambda} \phi$$

subject to

$$-\phi y_{i,k} + Y_k \lambda \geq 0$$

$$x_i - X \lambda \geq 0$$

$$\lambda \geq 0$$

and

$$[d^t(x_{t+1}, y_{t+1})]^{-1} = \max_{\phi, \lambda} \phi$$

subject to

$$-\phi y_{i,t+1} + Y_t \lambda \geq 0$$

$$x_{i,t+1} - X_t \lambda \geq 0$$

$$\lambda \geq 0$$

and

$$[d^{t+1}(x_t, y_t)]^{-1} = \max_{\phi, \lambda} \phi$$

subject to

$$-\phi y_{i,t} + Y_{t+1}\lambda \geq 0$$

$$x_{i,t} - X_{t+1}\lambda \geq 0$$

$$\lambda \geq 0$$

References

- Ali, A., Seiford, L., 1993. The mathematical programming approach to efficiency analysis. In: Fried, H., Lovell, C., Schmidt, S. (Eds.), *The Measurement of Productive Efficiency: Techniques and Applications*. Oxford University Press, New York, pp. 120–159.
- Association of University Technology Managers, 1997. Various years. AUTM Licensing Survey.
- Barfield, C., 1998. The irreplaceable federal role in funding basic scientific research, testimony before the Committee on Science. US House of Representatives.
- Charnes, A., Cooper, W., Lewin, A., Seiford, L., 1994. *Data Envelopment Analysis: Theory, Methodology and Applications*. Kluwer Academic Publishers, Boston.
- Cohen, W., Florida, R., Randazzese, L., Walsh, J., 1998. Industry and the academy: uneasy partners in the cause of technological advance. In: Noll, R. (Ed.), *Challenges to Research Universities*. Brookings Institute, Washington, DC, pp. 171–199.
- Fare, R., Grosskopf, S., Lovell, C., 1994. *Production Frontiers*. Cambridge University Press, New York.
- Fare, R., Grosskopf, S., Norris, M., Zhang, Z., 1994. Productivity growth, technical progress, and efficiency change in industrialized countries. *The American Economic Review* 84, 66–83.
- Geisler, E., Rubenstein, A., 1989. University–industry relations: a review of major issues. In: Link, A., Tasse, G. (Eds.), *Cooperative Research Development: The Industry–University–Government Relationship*. Kluwer Academic Publishers, Boston, pp. 43–64.
- Grey, D., Hetzner, W., Eveland, S., Gidley, T., 1986. NSF's industry–university cooperative research centers program and the innovation process: evaluation based lessons. In: Grey, D., Soloman, T., Hetzner, W. (Eds.), *Technological Innovation Strategies for a New Partnership*. North-Holland, New York, pp. 175–193.
- Grossman, J., Reid, P., Morgan, R., 2001. Contributions of academic research to industrial performance in five industry sectors. *Journal of Technology Transfer* 26, 143–152.
- Jensen, R., Thursby, M., 1999. Proofs and prototypes: a tale of university licensing. *American Economic Review*, in press.
- Lee, Y., 1996. Technology transfer and the research university: a search for the boundaries of university–industry collaboration. *Research Policy* 25, 843–863.
- Mansfield, E., Lee, J.-Y., 1996. The modern university: contributor to industrial innovation and recipient of industrial R&D support. *Research Policy* 25, 1047–1058.
- Morgan, R., Kannankutty, N., Strickland, D., 1997. Future directions for university-based research. *ASSEE PRISM* 6, 30–36.
- Mowery, D., Nelson, R., Sampat, B., Ziedonis, A., 2001. The growth of patenting and licensing by US universities: an assessment of the effects of the Bayh–Dole Act of 1980. *Research Policy* 30, 99–119.
- National Academy of Sciences, 1992. *The Government Role in Civilian Technology: Building a New Alliance*. National Academy Press, Washington, DC.
- National Research Council, 1995. In: Goldberger, M., Maher, B., Flattau, P. (Eds.), *Research Doctorate Programs in the United States*. National Academy Press, Washington, DC.
- National Science Board, 1993. *Science and Engineering Indicators*. National Science Board, Washington, DC.
- Parnes, M., Omenn, G., Brock, E., 2000. A case study of system complexity and regional approaches to technology transfer. *Journal of Technology Transfer*, in press.
- Public Policy Center for Stanford Research International, 1986. *The Higher Education–Economic Development Connection*. The American Association of State Colleges and Universities, Washington.
- Roessner, J., Wise, A., 1994. Industry perspectives on external sources of technology and technical information. *Policy Studies Journal* 22, 349–358.
- Seiford, L., Thrall, R., 1990. Recent developments in DEA: the mathematical programming approach to frontier analysis. *Journal of Econometrics* 46, 7–38.
- Thursby, J., Jensen, R., Thursby, M., 2001. Objectives, characteristics and outcomes of university licensing: a survey of major US universities. *Journal of Technology Transfer* 26, 59–72.
- Thursby, J., Thursby, M., 2000a. Who is selling the ivory tower? Sources of growth in university licensing. *Management Science*, in press.
- Thursby, J., Thursby, M., 2000b. Today's industry/university licensing environment: issues and problems as reported by participants.
- US General Accounting Office, 1998. *Technology Transfer: Administration of the Bayh–Dole Act by Research Universities*. GAO/RCED-98-126.